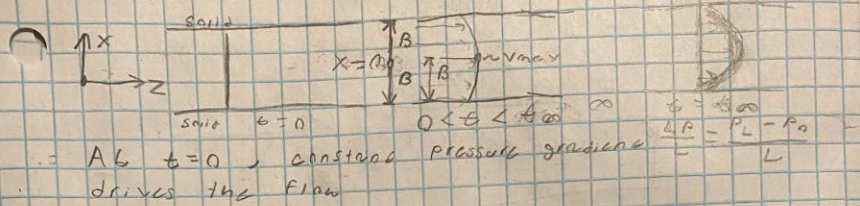


CHE 311 Problem set 4



- 1) Assuming continuity is satisfied, equation of motion? Normal = perpendicular
 μ, ρ both constants and Newtonian fluid
 $v_x = v_y = 0, v_z = v(x, t), \rho \frac{\Delta P}{L} = \frac{P_L - P_0}{L} = -\frac{\partial P}{\partial z}$
 A.P.P.B.G in cartesian:

$$x: \rho(0+0+0+0) = -0 + \mu[0+0+0] + 0$$

$$y: \rho(0) = \mu[0]$$

$$z: \rho \left(\frac{\partial v_z}{\partial t} \right) = -\frac{\partial P}{\partial z} + \mu \left[\frac{\partial^2 v_z}{\partial x^2} \right]$$

$$\text{Rearrange } z: \rho \frac{\partial v_z}{\partial t} = \mu \frac{\partial^2 v_z}{\partial x^2} - \frac{\partial P}{\partial z}$$

$$2) BC 1: v_z(x=B, t) = 0 \quad (1)$$

$$2: v_z(x=-B, t) = 0 \quad (2) \quad v_z(x=0, t) = v_{max}$$

$$IC: v_z(x, t=0) = 0$$

Obj: Get velocity profile $v_z(x)$

$$\rho \frac{\partial v_z}{\partial t} = \mu \frac{\partial^2 v_z}{\partial x^2} - \frac{\partial P}{\partial z} \quad \text{As } t \rightarrow \infty, \frac{\partial v_z}{\partial t} \rightarrow 0$$

$$\mu \frac{\partial^2 v_z}{\partial x^2} = + \frac{\partial P}{\partial z}$$

$$\mu \frac{\partial^2 v_z}{\partial x^2} = \frac{\Delta P}{L} \rightarrow \frac{\partial^2 v_z}{\partial x^2} = \frac{\Delta P}{\mu L} \rightarrow \text{Integrate}$$

$$\frac{\partial v_z}{\partial x} = \frac{\Delta P}{\mu L} x + C_1 \rightarrow \frac{\Delta P}{2\mu L} x^2 + C_1 x + C_2 = v_z$$

Using boundaries:

$$(1) 0 = \frac{\Delta P}{2\mu L} B^2 + C_1 B + C_2 = 0$$

$$(2) 0 = \frac{\Delta P}{2\mu L} B^2 + C_1 B + C_2 = 0$$

$$At x=0, \frac{\Delta P}{2\mu L} B^2 + 0 + 2C_2 = 0 \rightarrow C_2 = -\frac{\Delta P}{2\mu L} B^2$$

$$v_z = \frac{\Delta P}{2\mu L} x^2 + C_1 x - \frac{\Delta P}{2\mu L} B^2$$

Given $V_{max} = -\frac{B^2 \Delta P}{2\mu L}$

$$V_0 = \frac{\Delta P R^2}{2\mu L} - \frac{\Delta P B^2}{2\mu L}$$

$$= \frac{\Delta P B^2}{2\mu L} \left(\frac{x^2}{B^2} - 1 \right)$$

$$= -\frac{\Delta P B^2}{2\mu L} \left(1 - \left(\frac{x}{B} \right)^2 \right)$$

3) $\rho \frac{\partial v_z}{\partial t} = \mu \frac{\partial^2 v_z}{\partial x^2} - \frac{\Delta P}{L} \Rightarrow \frac{\partial \phi}{\partial \tau} = \frac{\partial^2 \phi}{\partial n^2} + 2$

$\phi = \frac{v_z}{V_{max}}, \tau = \frac{\mu t}{\rho B^2}, n = \frac{x}{B} = \frac{2\theta}{\theta_0}$

$\rho \frac{\partial v_z}{\partial t} = \mu \frac{\partial^2 v_z}{\partial x^2} - \frac{\Delta P}{L V_{max}} \Rightarrow \frac{\partial \phi}{\partial \tau} = \frac{\partial^2 \phi}{\partial n^2} - \frac{\Delta P}{L V_{max}}$

$$\frac{\partial \phi}{\partial \tau} = \frac{\partial \phi}{\partial \tau} \cdot \frac{\partial \tau}{\partial t} = \frac{\partial \phi}{\partial \tau} \left(\frac{\mu}{\rho B^2} \right) \quad \left(\frac{\partial \phi}{\partial \tau} \left(\frac{\mu}{\rho B^2} \right) \right)$$

$$\frac{\partial \phi}{\partial \tau} = \frac{\partial \phi}{\partial n} \cdot \frac{\partial n}{\partial x} \Rightarrow \frac{\partial^2 \phi}{\partial x^2} = \frac{\partial^2 \phi}{\partial n^2} \left(\frac{1}{B} \cdot \frac{1}{B} \right) \quad \left(\mu \frac{\partial^2 \phi}{\partial n^2} \frac{1}{B^2} \right)$$

$$\frac{\partial \phi}{\partial \tau} \left(\frac{\mu}{\rho B^2} \right) = \mu \frac{\partial^2 \phi}{\partial n^2} \frac{1}{B^2} - \frac{\Delta P}{L V_{max}}$$

$$\frac{\partial \phi}{\partial \tau} = \frac{\partial^2 \phi}{\partial n^2} - \frac{\Delta P}{L V_{max}} \frac{B^2}{\mu} + \frac{2\mu B}{B^2 \Delta P} \cdot \frac{\Delta P}{L} \cdot \frac{B^2}{\mu}$$

$$\left[\frac{\partial \phi}{\partial \tau} = \frac{\partial^2 \phi}{\partial n^2} + 2 \right]$$

4) Get the steady state velocity profile in the form $\phi = 1 - n^2$

$$\frac{\partial \phi}{\partial \tau} = \frac{\partial^2 \phi}{\partial n^2} + 2 \quad \phi(n, \tau) \rightarrow \phi_a(n)$$

BC 1: $\phi(n=0, \tau) = 0$ $\phi(n=\pm 1, \tau) = 0$ ②

BC 1: $\phi(n=1, \tau) = 0$ ①

IC: $\phi(n, \tau=0) = 0$

$$0 = \frac{\partial^2 \phi_a}{\partial n^2} + 2 \rightarrow -2 = \frac{\partial^2 \phi_a}{\partial n^2}$$

$$-2n + c_1 = \frac{\partial \phi_a}{\partial n} \rightarrow -\frac{n^2}{2} + c_1 n + c_2 = \phi_a$$

$$\textcircled{1} (-1)^2 - c_1 + c_2 = 0 \quad c_2 = 0$$

$$\textcircled{2} -1^2 + c_1 + c_2 = 0$$

$$\frac{2}{2} + 2c_2 = 0 \rightarrow c_2 = -1$$

$$\boxed{\phi_a = -n^2 + 1}$$

5) Initial and two boundary conditions

$$\phi(n, \tau=0) = 0$$

$$\phi(n=-1, \tau) = 0 \quad \text{Done?}$$

$$\phi(n=1, \tau) = 0$$

6) Substitute $\phi(n, \tau) = \sum_{n=1}^{\infty} c_n(\tau) \cos\left(\frac{2n-1}{2} \pi n\right)$

in the partial differential equation

$$\frac{\partial \phi}{\partial \tau} = \frac{\partial^2 \phi}{\partial n^2} + 2$$

Obj. 1: derive an ODE for $c_n(\tau)$

$$\frac{\partial \phi}{\partial \tau} = \sum_{n=1}^{\infty} c_n(\tau) \left[-\sin\left(\frac{2n-1}{2} \pi n\right) \right] \cdot \frac{2n-1}{2} \pi$$

$$\frac{\partial^2 \phi}{\partial n^2}$$

$$\frac{\partial \phi}{\partial \tau}$$

$$\frac{\partial^2 \phi}{\partial n^2} = \sum_{n=1}^{\infty} c_n(\tau) \left[-\cos\left(\frac{2n-1}{2} \pi n\right) \right] \cdot \frac{2n-1}{2} \pi \cdot \frac{2n-1}{2} \pi$$

$$\frac{\partial \phi}{\partial \tau} = \frac{\partial \phi}{\partial c_n} \cdot \frac{\partial c_n}{\partial \tau} = \frac{\partial \phi}{\partial c_n} \cdot \frac{\partial c_n}{\partial \tau} = \frac{\partial \phi}{\partial c_n} \cdot \frac{\partial c_n}{\partial \tau}$$

$$\frac{\partial^2 \psi}{\partial n^2} = \sum_{v=1}^{\infty} c_v(\gamma) \left(-\cos \frac{2v-1}{2} \pi n \right) \cdot \frac{(2v-1)^2}{4} \pi^2$$

$$\frac{\partial \psi}{\partial \tau} = \frac{\partial \psi}{\partial b} \cdot \frac{\partial b}{\partial \tau} = \sum_{v=1}^{\infty} c_v \cos \left(\frac{2v-1}{2} \pi n \right)$$

$$\cos \left(\frac{2v}{2} \pi n - \frac{1}{2} \pi n \right) = \cos a \cos B + \sin a \sin B$$

$$\frac{2a}{\pi} \int_0^{\frac{\pi}{2}} dx \cos(nax) \cos(max) = \frac{a}{\pi} \int_0^{\frac{\pi}{2}} dx [\cos[(n-m)x] + \cos[(n+m)x]]$$

$$\frac{2a}{\pi} \int_0^{\frac{\pi}{2}} dx \sin(nax) \sin(max) = \frac{a}{\pi} \int_0^{\frac{\pi}{2}} dx [\cos[(n-m)x] - \cos[(n+m)x]]$$

$$\frac{\partial c_v}{\partial b} = -c_v(\gamma) \cdot \frac{(2v-1)^2}{4} \pi^2 + \frac{2}{\sum \cos \left(\frac{2v-1}{2} \pi n \right)}$$

$$c_v = -c_v(\gamma) \cdot \frac{(2v-1)^2}{4} \pi^2 b + \frac{2}{\sum \cos \left(\frac{2v-1}{2} \pi n \right)} + c_2$$

$$+ \frac{2}{\cos a \cos B + \sin a \sin B} + c_2$$

I give up